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337614 (37)

BE (6th Semester)

Examination, April-May, 2014

Branch : Mechanical

MACHINE TOOL TECHNOLOGY

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) All questions are compulsory.

(ii) Attempt questions according to internal choice.

(iii) Part (a) is compulsory and carries 2 marks each in all questions.

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(iv) Solve any two parts from (b), (c) & (d) of each question.

(v) Draw diagrams neatly wherever necessary.

Q. 1. (a) Define tool signature. 2

(b) Explain the different cutting tool materials and explain their characteristics. 7

(c) What are the assumptions made in Merchant's Theory of metal cutting ? How will you calculate stress and strain in the chip ? Draw a neat sketch. 7

(d) List the different types of tool angle specification system. Explain any one. 7

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Q. 2. (a) What do you understand by the term "machinability". 2

(b) What is the effect of various parameters on tool life. Derive Taylor's tool life relationship involving cutting parameters. 7

(c) Explain the mechanism of tool wear. Write the differences between face wear and flank wear. 7

(d) Describe the working of a tool force dynamometer for measuring cutting forces during turning operation. 7

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Q. 3. (a) Write two uses of renforcing stiffner in lathe beds. 2

(b) Discuss the functions and requirements of machine tool structure. 7

(c) Explain the analysis of forces under headstock using a neat sketch. 7

(d) Write short notes on guide ways. 7

Q. 4. (a) Define deviation diagram. Explain with a simple example. 2

(b) What is productivity loss ? How can one control it ? Explain briefly with the help of sketches.

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(c) Draw the best possible structural diagram of a machine tool speed gear box having speeds ranging from 45 rpm to 710 rpm with $\phi = 1.41$. If the motor runs at 1440 rpm. Determine the number of teeth on all gears. 7

Q. 5. (a) What is norton drive ? 2

(b) Explain the procedure for designing kinematic arrangement of apron mechanism for cutting metric threads on lathe. 7

(c) Explain "machine tool maintenance" in detail. 7

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(d) Briefly explain the acceptance test carried out on radial drilling machine. 7

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